```
from google.colab import files
uploaded = files.upload()
```

Choose files mnist_train.csv.zip
mnist_train.csv.zip(application/x-zip-compressed) - 13692401 bytes, last modified: 23/03/2026 - 100% done
Saving mnist_train.csv.zip to mnist_train.csv.zip

```
import pandas as pd
import zipfile

# Unzip the uploaded file
with zipfile.ZipFile('mnist_train.csv.zip', 'r') as zip_ref:
    zip_ref.extractall('.')

data = pd.read_csv("mnist_train.csv")
print(data.shape)
data.head()
```

(60000, 785)

	label	1x1	1x2	1x3	1x4	1x5	1x6	1x7	1x8	1x9	...	28x19	28x20	28x21	28x22	28x23	28x24	28x25	28x26	28x27	28x28
0	5	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
2	4	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
3	1	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0
4	9	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	0

5 rows × 785 columns

```
X = data.iloc[:, 1:]
y = data.iloc[:, 0]
```

```
X = X / 255.0
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

lr = LogisticRegression(max_iter=1000)
lr.fit(X_train, y_train)
y_pred_lr = lr.predict(X_test)

print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred_lr))
```

```
Logistic Regression Accuracy: 0.9180833333333334
```

```
from sklearn.neighbors import KNeighborsClassifier

knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
y_pred_knn = knn.predict(X_test)

print("KNN Accuracy:", accuracy_score(y_test, y_pred_knn))
```

```
KNN Accuracy: 0.9715
```

```
from sklearn.svm import SVC

svm = SVC()
svm.fit(X_train, y_train)
y_pred_svm = svm.predict(X_test)

print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
```

```
SVM Accuracy: 0.9768333333333333
```

```
from sklearn.tree import DecisionTreeClassifier

dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)
y_pred_dt = dt.predict(X_test)

print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))
```

```
Decision Tree Accuracy: 0.8641666666666666
```